25AE-0270

Development of a Rule-Based Lidar Lane-keeping Assistance System for Autonomous Vehicles Applications

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Abstract

Lane-keeping is critical for SAE Level 3+ autonomous vehicles, requiring rigorous validation and end-to-end interpretability. All recently U.S.-approved level 3 vehicles are equipped with lidar, likely for accelerating active safety. Lidar offers direct distance measurements, allowing rule-based algorithms compared to camera-based methods, which rely on statistical methods for perception. Furthermore, lidar can support a more comprehensive and detailed approach to studying lane-keeping. This paper proposes a module perceiving oncoming vehicle behavior, as part of a larger behavior-tree structure for adaptive lane-keeping using data from a lidar sensor. The complete behavior tree would include road curvature, speed limits, road types (rural, urban, interstate), and the proximity of objects or humans to lane markings. It also accounts for the lane-keeping behavior, type of adjacent and opposing vehicles, lane occlusion, and weather conditions. The algorithm was evaluated using experimental lidar data collected from driving around Georgia Southern's campus on one of the behavior tree's most intensive inputs: oncoming vehicle lane-keeping behavior in two-way, two-lane highways with no physical barriers. Preliminary results include demonstrating one behavior-tree module recognizing an oncoming vehicle's lane-keeping ability, showing a promising future for interpretable algorithms when using lidar. Existing and novel methods were combined to acquire behavior metrics: Distance to Lane Marking (DTLM), trajectory prediction error (pE), the relative distance between ego- and target vehicles, predicted dividing lane crossings and the number of vehicle points tracked (NoP).

Introduction

Advanced Driver Assistance Systems (ADAS) have been a significant step toward certified autonomy in the automotive industry with promising advances in vehicle safety. Based on a study of U.S. crashes in Crash Investigation Sampling System (CISS) between 2014 and 2016, vehicles equipped with Lane Keeping Assistance Systems (LKAS) reduced the overall number of single-vehicle and head-on collisions by 60% ± 16%, compared to vehicles not equipped with any ADASs [1]. The recent availability of SAE L2 [2] vehicles, featuring ADAS such as Forward Collision Avoidance/Warning (FCA/W) and Lane Keeping Assistance/Warning (LKA/W) have shown promising improvement in roadway safety globally. Research in ADAS can gradually pave the way for certified SAE Level 3+ vehicles as each module such as perception, decision-making, and control is improved or better understood in terms of its performance and reliability limits.

The primary objective of ADAS is to assist drivers in targeted driving tasks including emergency responses for lane-keeping and emergency braking [3, 4, 5]. These functions are expected to work reliably, especially with SAE L3 vehicles defined as conditionally autonomous. Finding a balance between performance and interpretability is a common challenge when selecting statistical machine learning (ML) models over traditional modeling for notoriously complex autonomous driving systems. The recent successes of research on ML-based perception, decision-making, and control show a likely requirement for such models approaching SAE L5 vehicles. However given the conditional nature of L3, and rich information available, traditional rule-based algorithms such as behavior trees would likely be adequate. Recently, L3 vehicles have become certified for sale in several countries. A key difference between L2 vehicles and the certified L3 vehicles so far is the addition of lidar. This addition in a leap from L2 to L3 shows the reliability value that the lidar offers to the industry.

An LKAS addresses a key component of AD across all SAE driving autonomy levels: maintaining a lane. LKA involves active steering controls upon the detection or prediction of a lane departure, while LDW solely informs the driver of an imminent lane departure. Furthermore, two primary types of LKAS exist Type I involves a steering correction once an imminent lane departure is detected, and Type II involves active lane-centering assistance [5]. In any LKAS, two primary functions exist. Firstly, the lane markings in front of the vehicle are perceived. Secondly, imminent lane departures are either detected or predicted to produce an active steering correction. The work presented in this paper builds on a simple near-range, lidar-based lane marking detection method similar to that presented in [6]. Our scope builds on the aforementioned works by developing an interpretable behavior-tree approach to achieve an improved lidar-powered lane-keeping method. The specific results presented in this paper showcase one module of our proposed behavior-tree framework: an ego lane-keeping adjustment based on the lane-keeping behavior of an oncoming vehicle.

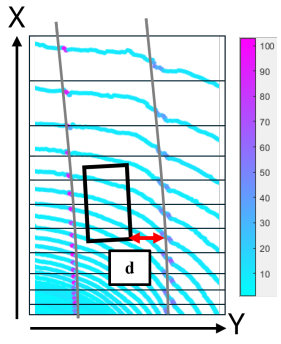


Figure 2: Opposing Vehicle Lane-Keeping Behavior Tracking. This figure uses experimental data obtained by the authors

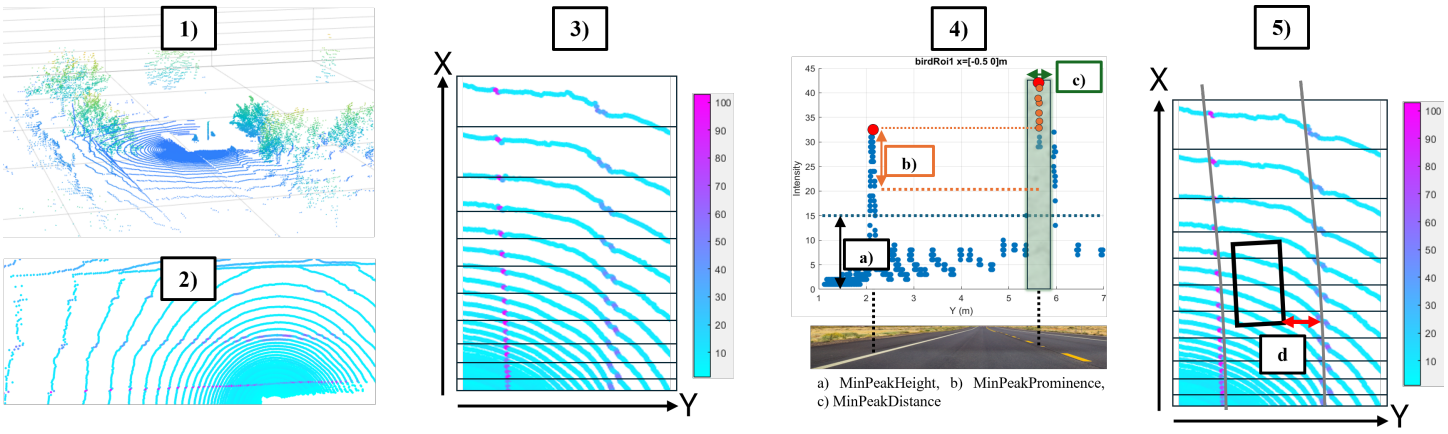


Figure 1: Lidar-Based Lane and Oncoming Vehicle Behavior Quantification - 1) Raw 3D Point Cloud; 2) Birds-Eye-View of Roadway with Reflectivity Returns; 3) ROI Binning; 4) Reflectivity Spike Localization; 5) Determine Distance of Front Driver-Side Corner of Oncoming Vehicle's Bounding Box to Poly-Fit of Lane Detections. These figures all use experimental data obtained by the authors

Autonomous driving is deeply intertwined with deep learning algorithms thus enabling the necessary reaction time. Yet the readability and comprehensibility of these models are low. Emerging from the video game industry in the early 2000s, Behavior Trees (BT) quickly became popular in robotics and further evolved into the autonomous driving industry—their key strength sources in their directed graph with a tree structure [7]. A BT has one topmost node called the root from where the entire tree structure originates. The two types of nodes are composite nodes, analog to branches of a tree that are responsible for the logic of the tree execution, and leaves, analog to the ends of the branches that are the acting nodes. Underneath it follows the execution logic of a decision tree (nested if-else structure) with the crucial advantage that complex structures are summed up as one composite node in the BT and therefore streamline debugging and predictability of the code.

During the execution, the state of a node can either be Success, Failure, or Running and their priority level decreases from left to right as the tree is explored in a depth-first manner starting from the root node [7]. To make the BT reactive in the case of a node returning "running" in between ticks and therefore blocking the BT, the overlaying sequence or selector can be configured without memory ensuring that the high-priority nodes on the left are checked again with every tick ensuring accurate logic for quickly changing situations.

The success of recent autonomous driving behavior projects based on BTs proves this concept is ideal for it. This paper builds on those results and uses them for centerline perception from real-world 3D-lidar driving data, improving transparency compared to conventional camera-based deep learning models.

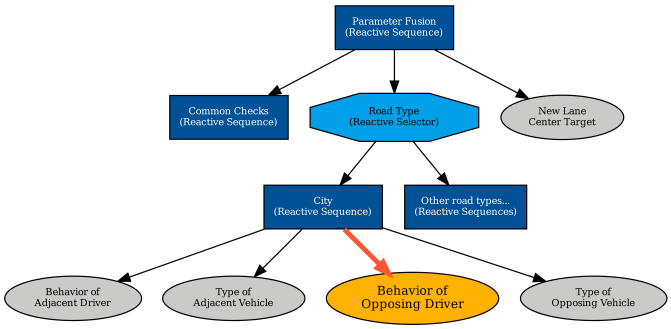


Figure 3: Overlaying Behavior Tree

Related Works

Today three standards exist regulating the performance of LKAS: UN ECE R79 [8], ISO 11270 [9], and EURO NCAP [10]. These standards have slightly varying test scenarios and passing criteria, however, all address the adequate performance of LKAS [5]. L2 vehicles always require the driver's full attention, and therefore lack specific safety and reliability standards. Lane-marking detection has primarily been deployed with vision-based systems, such as works well summarized in [11]. While several works use lidar to detect lanes [6, 11, 12, 13, 14, 15, 16, 17], to the best of the authors' knowledge, no certified deployments of these methods exist today. Though a fusion is possible considering the handful of vehicles offering lidar in their ADAS packages, this is not likely as L1-L2 vehicles so far have almost entirely been equipped with only camera, radar, and ultrasonic sensors. Driver behavior while maintaining a lane is well-studied [18, 19, 20], however to the best of the authors' knowledge, no work exists regarding an adaptive lane-keeping method based on the behavior of oncoming vehicles. A similar problem of avoiding collisions with oncoming vehicles is typically approached as a reactive "collision avoidance" problem compared to our "preventative lane-keeping" approach and is thoroughly studied in [21, 22, 23, 24, 25].

Lidar sensors provide point cloud measurements of their environment where each point corresponds to returned laser emissions. Contrary to RGB cameras, lidar sensors are active as their laser diodes emit and then receive near-infrared (NIR) light [26]. While this makes lidar resilient to low-light conditions [27], dust, rain, and fog act as a reflecting and refracting barrier causing false positives [28, 29, 30, 31]. Dual-return is a lidar technology that recognizes multiple significant returns for the same laser emission, signifying light travel through a low-refractive water particle [32]. Dual-return is implemented on all state-of-the-art automotive lidar sensors available and has been shown to reduce the effects of rain and fog by sensing objects behind the translucent particles. ML-based methods have been proposed to improve dual-return [33] and no promise exists to combat the effects of opaque particles such as dust. Nevertheless, research on camera-lidar-radar fusion suggests that lidar can play a key role in AD perception, especially for low-light conditions [34, 35, 36].

A review paper on the use of AI for AD [37] mentions some key deficiencies of ML models, applicable to LKA ADAS: A small change to camera images may cause ML-based ADAS systems to fail in lane detection and segmentation [38]. Generating accurate annotations of lanes for ML model training is time-consuming [39]. To verify catastrophic failure rates of ML models, a significant 109 hours of vehicle operation testing is necessary [40] with repetitions for statistical significance [41]. Vriesman et al. focused on the life-critical issue of



Figure 4: Data Capturing Map

pre-crash scenarios, employing both lidar and camera sensors to predict potential crash scenarios [42]. Their work has shown that multi-sensor approaches can predict crash scenarios with higher accuracy than single-sensor methods. This is particularly important for safety-critical applications, especially in autonomous driving where timely and accurate pre-crash detection can save lives [43].

Methodology

The demonstrated ADAS module of a preventative ego lane-keeping adjustment is one based on the lane-keeping behavior of an oncoming vehicle. In this section, all algorithms utilized to derive the results presented are described and were implemented on MATLAB 2024b. The proposed method's perception functions were applied to a lidar driving dataset of nine detected scenarios taken on Georgia Southern University's campus roadways using an Ouster OS0 Rev7 128-channel lidar sensor.

Lane Detection and Distance to Lane-Marking (DTLM)

This module begins by perceiving the lane markings in a specified region of interest (ROI) based on methods described by [44, 45]. Here, the ground point cloud is first separated from the whole point cloud to detect intensity spikes in consecutive 2D bins of the ROI. The lane marking detections represented by the intensity spikes are fit to a second-order polynomial in Eqn. (1). Then, the distance of an oncoming vehicle's front left corner to its left lane marking is determined by performing a "QuickHull" [46] minimum search of polynomial distances to 100 discrete points in the ranges defined in Eqn. (3).

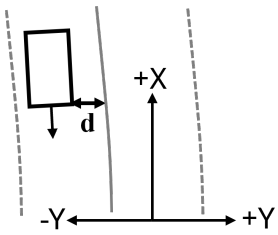


Figure 5: Lane-Keeping Distance Geometries

$$y = a_2x^2 + a_1x + a_0 \tag{1}$$

The vehicle's bounding box is determined by its position and dimensions as detected in a lidar detection and tracking method described in the work of [47]. This method gives the real-time information of an oncoming vehicle's bounding box seen in matrix (2).

$$x_{ct} = [x\ \dot{x}\ y\ \dot{y}\ \theta\ z\ \dot{z}\ \theta\ l\ w\ h] \tag{2}$$

The length of tracked oncoming vehicles is used to define a range where points of the dividing lane marking are generated to find the minimum distance between the lane marking poly-fit and the oncoming vehicle's query point, where:

$$x_{\min} = x - \frac{3l}{2}, \quad x_{\max} = x + \frac{3l}{2} \tag{3}$$

To find the closest point on the polynomial to the vehicle's front driver-side corner, we adjust the vehicle's position by subtracting half the length l from its x-coordinate and adding half the width w to its y-coordinate, resulting in the query point:

$$\text{QP} = (x - \frac{l}{2}, y + \frac{w}{2}) \tag{4}$$

The nearest point on the polynomial to the query point is determined using the Quickhull algorithm, as described by [46]. The algorithm efficiently computes the nearest point (NP) to the query point:

$$\text{NP} = \text{Quickhull}(P, QP) \tag{5}$$

Finally, the Euclidean distance between NP and QP, quantifying the opposing vehicle's lane-keeping behavior is computed as:

$$DTLM = \sqrt{(x_p - x)^2 + \left(y_p - \left(y + \frac{w}{2}\right)\right)^2} \tag{6}$$

The geometries described in this methodology are visualized in Fig. 5 and the full lidar data processing pipeline is summarized in Fig. 1.

Oncoming Vehicle Number of Points (NoP)

Tracking an oncoming vehicle's NoP within its bounding box provides some insight on the quality of the vehicle tracking. Furthermore, lower NoP while tracking a vehicle can provide information on the vehicle, such as low-reflective, high-absorptive, or small-dimension vehicle known to induce less accurate tracking [48]. NoP is a count of lidar points found within the limits of the oncoming vehicle's tracked bounding box. To achieve this at each frame i, the bounding box's position and dimensions in x_{ct} (2) are processed in (7), (8), (9), to generate an ROI in (10).

$$x_{\min}(i) = x_{\text{vehicle}}(i) - \frac{l(i)}{2}, \quad x_{\max}(i) = x_{\text{vehicle}}(i) + \frac{l(i)}{2} \tag{7}$$

$$y_{\min}(i) = y_{\text{vehicle}}(i) - \frac{w(i)}{2}, \quad y_{\max}(i) = y_{\text{vehicle}}(i) + \frac{w(i)}{2} \tag{8}$$

$$z_{\min}(i) = z_{\text{vehicle}}(i) - \frac{h(i)}{2}, \quad z_{\max}(i) = z_{\text{vehicle}}(i) + \frac{h(i)}{2} \tag{9}$$

$$\text{ROI}(i) = [x_{\min}(i), x_{\max}(i), y_{\min}(i), y_{\max}(i), z_{\min}(i), z_{\max}(i)] \tag{10}$$

A searching algorithm $findPointsInROI$ provides indices of points found within the ROI in the obstacle-segmented pointcloud $obsPC(i)$. These indices are then counted to achieve $NoP(i)$ in function (11).

$$\text{NoP}(i) = \text{count}(\text{findPointsInROI}(\text{obsPC(i)}, \text{ROI(i)})) \tag{11}$$

Oncoming Vehicle Trajectory Prediction

While DTLM provides a direct estimation of the oncoming vehicle's lane-keeping real-time behavior, predicting imminent vehicle trajectories can provide such a system with further behavior insight. This is done using, $\dot{x}$, which represents the oncoming vehicle's velocity in the longitudinal direction, $\dot{y}$, which represents the velocity in the lateral direction, and $\dot{\theta}$, which represents the yaw rate of change.

Using kinematic equations, the oncoming vehicle's predicted position (x_{traj}, y_{traj}) and yaw (θ_{traj}) is determined at each point in time (k). The full-time domain for predictions is defined in Eqn. (12).

$$\Delta t = 0.01\text{ s}, \quad T = 0.1\text{ s}, \quad \text{timetraj} = [0, \Delta t, 2\Delta t, \ldots, T] \tag{12}$$

The final prediction time T was chosen as 0.1s, to align with the lidar's data rate for a trajectory error evaluation. For the time domain $timetraj$ in Eqn. (12), the oncoming vehicle's trajectory is evaluated using Eqns. (13), (14), (15).

$$x_{\text{traj}}(k) = x_{\text{traj}}(k-1) + \dot{x}\Delta t \tag{13}$$

$$y_{\text{traj}}(k) = y_{\text{traj}}(k-1) + \dot{y}\Delta t \tag{14}$$

$$\theta_{\text{traj}}(k) = \theta_{\text{traj}}(k-1) + \dot{\theta}\Delta t \tag{15}$$

Using the oncoming vehicle's states $k = 1$ and $k = N$, at each tracked frame i, $initial$ and $prediction$ are stored.

$$\text{prediction}(i) = [x_{\text{traj}}(N), y_{\text{traj}}(N), \theta_{\text{traj}}(N)] \tag{16}$$

$$\text{intial}(i) = [x_{\text{traj}}(1), y_{\text{traj}}(1), \theta_{\text{traj}}(1)] \tag{17}$$

The difference of the values stored in matrices (17) and (16), at each frame i is the prediction error pE defined in Eqn. (18).

$$\text{pE}(i) = \text{intial}(i) - \text{prediction}(i) \tag{18}$$

Evaluating the prediction error is important as one can predict and prepare for critical scenarios where an oncoming vehicle is on track to cross the lane marking. Predicted Crossings is a count metric evaluated on all scenarios to determine any false positive predictions, as no actual crossing occurred in any of the presented scenarios.

Results

The results include an application of NoP, pE, and DTLM metrics on nine recorded scenarios, summarized in four plots and two tables. Each plot's independent variable is time in a "t minus" configuration where t = 0ms represents the point at which the oncoming vehicle's front end has passed the ego vehicle's back end. After t = 0s, no risk of a head-on collision is anticipated given the nature of the problem. All plots begin at t = -1500ms, where the earliest vehicle detection occurred (Scen. 6). All scenarios will be briefly described and outlying phenomena, especially visible in pE and DTLM, are described in further detail.

Scenario 1-3: The ego vehicle encountered a single-lane marking separating the two lanes. In these scenarios, the oncoming vehicle was a compact SUV (Scenario 1), a pickup truck (Scenario 2), and a compact sedan combined with a low-intensity lane marking ().

Scenario 4-8: The ego vehicle faced a double lane marking separating the lanes. The oncoming vehicles were a compact SUV (Scenario 4), a compact sedan (Scenarios 5, and 8), a compact sedan combined with low-intensity lane markings (Scenario 7), and a pickup truck (Scenario 6).

Scenario 9: A unique situation was introduced where the ego vehicle passed a stationary compact pickup truck located in a median lane. This scenario incorporated both single and double-lane markings.

The scenarios represent this work's problem statement: passing oncoming vehicles on a two-way, two-lane highway with no physical barrier. An intuitive way to visualize all scenarios is in Fig. 6, where the oncoming vehicle's relative distance to the sensor over time is shown. The relative velocity of the oncoming vehicle to the ego vehicle is determined using a least-squares line-fit of these distances and displayed in Table 1 and Table 2. A slight convergence of distance is visible approaching t = 0, as the oncoming vehicle passes the ego vehicle.

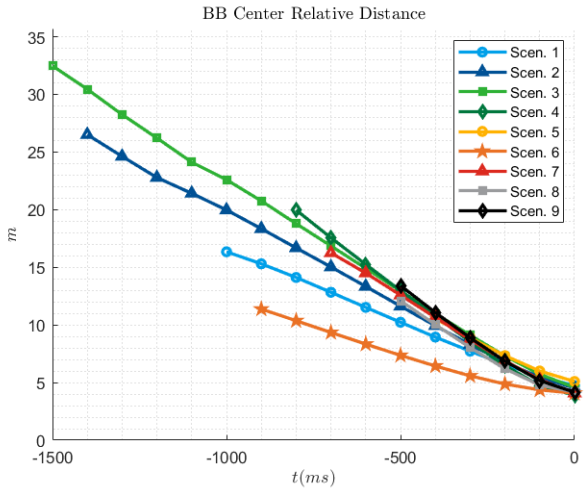


Figure 6: Oncoming Vehicle Distance (d rel.=$\sqrt{(x)^2 + (y)^2}$)

As 3D lidar sensors emit and receive light pulses at fixed angular resolutions, the number of points returned on any perceived vehicle will increase exponentially as it approaches the sensor. This is visible for all scenarios in Fig. 7. The NoP is seen to break its exponential increase as it approaches t = 0 because the vehicle continues to be tracked while next to the ego vehicle. Scenario 5's late detection can be attributed to a low initial NoP, while Scenario 6's higher NoP is likely due to the larger size of the oncoming vehicle.

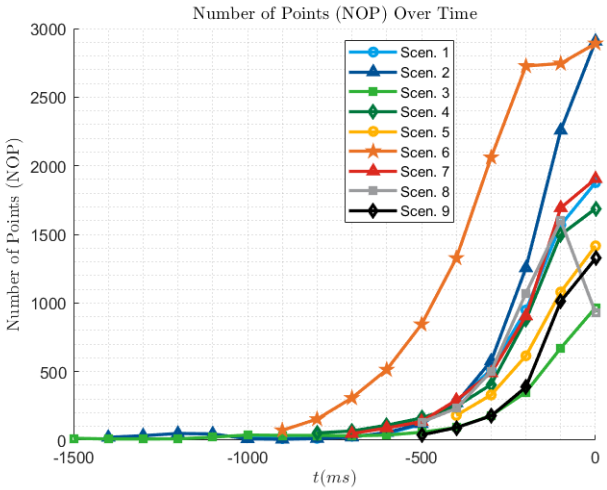


Figure 7: Number of Points (NoP) of Tracked Oncoming Vehicle

The anomalous spike in Scenario 1's NoP between t = -800ms and -500ms in Fig. 8 is a result of inaccurate bounding box yaw angle estimation. The reason for the high error in at -1100ms remains unclear.

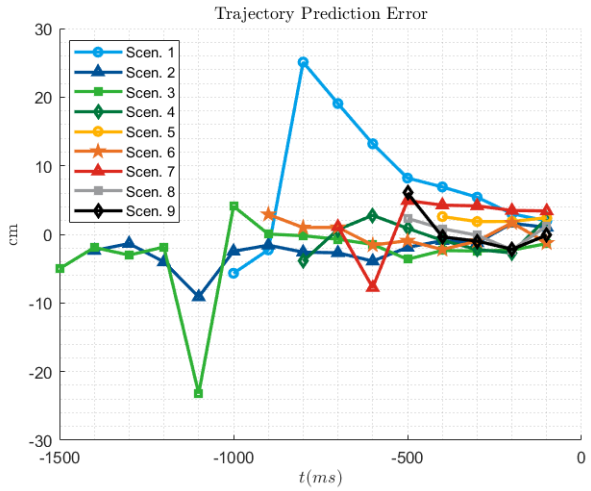


Figure 8: Trajectory Prediction Error (pE)

Scenario 3's spike at t = -1300ms in Fig. 9 is believed to be caused by an extreme lane-detection error caused by the low-intensity lane marking. Similarly, the spike in Scenario 7 at t = 0ms is due to the lane-detection algorithm mistakenly considering points from the oncoming vehicle as lane-marking points.

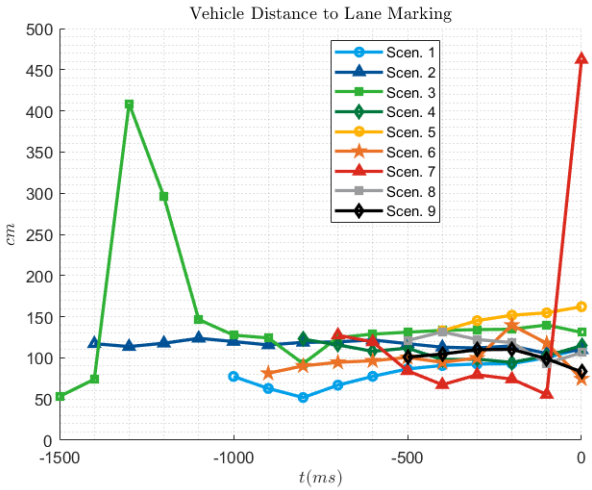


Figure 9: Distance to Dividing Lane Marking (DTLM)

The data presented in Tables 1 and 2 summarize the key performance metrics of the system across nine scenarios. Table 1 outlines the system's performance in terms of DTLM, showing significant variation in DTLM max values, particularly in Scenarios 3 and 7, where the DTLM max reached 4.09 and 4.62 meters, respectively, due to low-intensity lane markings. In contrast, Scenario 5 demonstrated more stable lane-keeping, with a DTLM mean of 1.49 meters and minimal deviation. Table 2 provides insights into trajectory prediction errors (Pred. Error Mean) and the maximum number of lidar points (NoP) detected within the oncoming vehicle's bounding box. The prediction errors remained minimal across all scenarios, with Scenario 1 showing a slightly higher error of 0.09 meters. The NoP values, which reflect the accuracy of vehicle detection, were generally higher for larger vehicles, with Scenario 2 recording the highest NoP at 2,907 points. This data collectively highlights the system's ability to track and predict vehicle behavior with a high degree of accuracy, although performance is dependent on the perception of the lane markings.

Conclusion

This research highlights the potential of high-resolution lidar sensors in lane-keeping assistance systems for autonomous vehicles. The developed rule-based algorithm, focused on lane detection and oncoming vehicle behavior, successfully demonstrated lidar's ability to provide crucial, interpretable data to enhance lane-keeping safety. The experimental results, based on real-world driving data, showcase the system's effectiveness in tracking lane markings and predicting oncoming vehicle trajectories.

DTLM determination seemed stable, particularly in scenarios where lane markings were clearly defined and well-maintained. For instance, Scenario 5, which involved passing an oncoming sedan with double lane markings, yielded a stable DTLM with an average distance of 1.49 meters and a low deviation of 0.11 meters, reinforcing the system's reliability under optimal conditions. However, challenges were seen in , where low-intensity lane markings caused significant variations in DTLM (ranging from 0.53 to 4.09 meters). This shows the system's sensitivity to poorly maintained lanes.

The NoP data further validated the system's ability to detect and track oncoming vehicles reliably. In all scenarios, NoP increased rapidly as oncoming vehicles approached, reflecting precise vehicle detection. Although anomalies, such as the yaw angle error in Scenario 1, were noted, it did not significantly affect the system's overall performance. Additionally, the system's ability to maintain low pE reinforces this method's reliability.

Future work will focus on extending the system's capabilities to more complex environments, such as urban settings and highways with multiple lanes. This will include integrating additional modules into the behavior tree framework and refining algorithms to enhance robustness under various weather, lighting, and road conditions. Future work will also involve improving the lane marking perception as several scenarios in the dataset were omitted for hard-to-detect lane markings. As lidar technology itself continues to develop, improvements in detection range and accuracy are expected, further advancing the system's overall performance and value in vehicle autonomy applications.

Table 1: Summary of Results Tabulated (m) - Part 1

Scenario	V rel. mean	V rel. R2	DTLM min	DTLM max	DTLM mean	DTLM std. dev.	1st Detection Distance
1	12.06	0.9978	0.52	1.11	0.83	0.17	16.35
2	16.01	0.9986	1.05	1.24	1.16	0.05	26.51
3	18.87	0.9988	0.53	4.09	1.49	0.86	32.46
4	20.78	0.9914	0.95	1.23	1.07	0.10	19.96
5	13.81	0.9881	1.33	1.62	1.49	0.11	10.56
6	8.44	0.9827	0.75	1.40	0.99	0.18	11.37
7	17.91	0.9966	0.56	4.62	1.34	1.35	16.25
8	16.33	0.9746	0.93	1.32	1.16	0.14	12.10
9	18.85	0.9849	0.83	1.10	1.01	0.10	13.41

Table 2: Summary of Results Tabulated (m) or (#) if stated - Part 2

Scenario	Pred. Error Mean	Pred. Error Std. Dev.	Pred. Min Error	Pred. Max Error	Max NoP (#)	Pred. Crossings (#)
1	0.09	0.09	0.02	0.25	1877	0
2	0.03	0.03	0.01	0.09	2907	0
3	0.04	0.06	0.00	0.23	967	0
4	0.02	0.02	0.01	0.04	1688	0
5	0.02	0.00	0.02	0.03	1416	0
6	0.02	0.02	0.01	0.03	2891	0
7	0.04	0.04	0.01	0.08	1905	0
8	0.01	0.02	0.00	0.02	1600	0
9	0.02	0.03	0.00	0.06	1325	0

Definitions, Acronyms, Abbreviations

ADAS	Advanced Driver Assistance Systems
AD	Autonomous Driving
BT	Behavior Tree
CISS	Crash Investigation Sampling System
DTLM	Distance to Lane Marking
FCA	Forward Collision Avoidance
FCW	Forward Collision Waring
Lidar	Light Detection and Ranging
LKA	Lane Keeping Assistance
LKAS	Lane Keeping Assistance System
LDW	Lane Departure Warning
ML	Machine Learning
NHTSA	National Highway Traffic Safety Administration
NIR	Near-Infrared Light
NoP	Number of Points Detected
NP	Nearest Point
pE	Predicted Error
QP	Query Point
ROI	Region of Interest
ToF	Time of Flight

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